

CSC 361

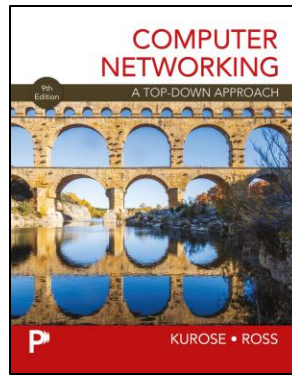
Computer Networks

Application Layer

(Chapter 2)

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Application Layer: Overview (4 of 7)

- Principles of network applications
- Web and HTTP
- E-mail, SMTP, IMAP
- The Domain Name System DNS
- video streaming and content distribution networks
- socket programming with UDP and TCP



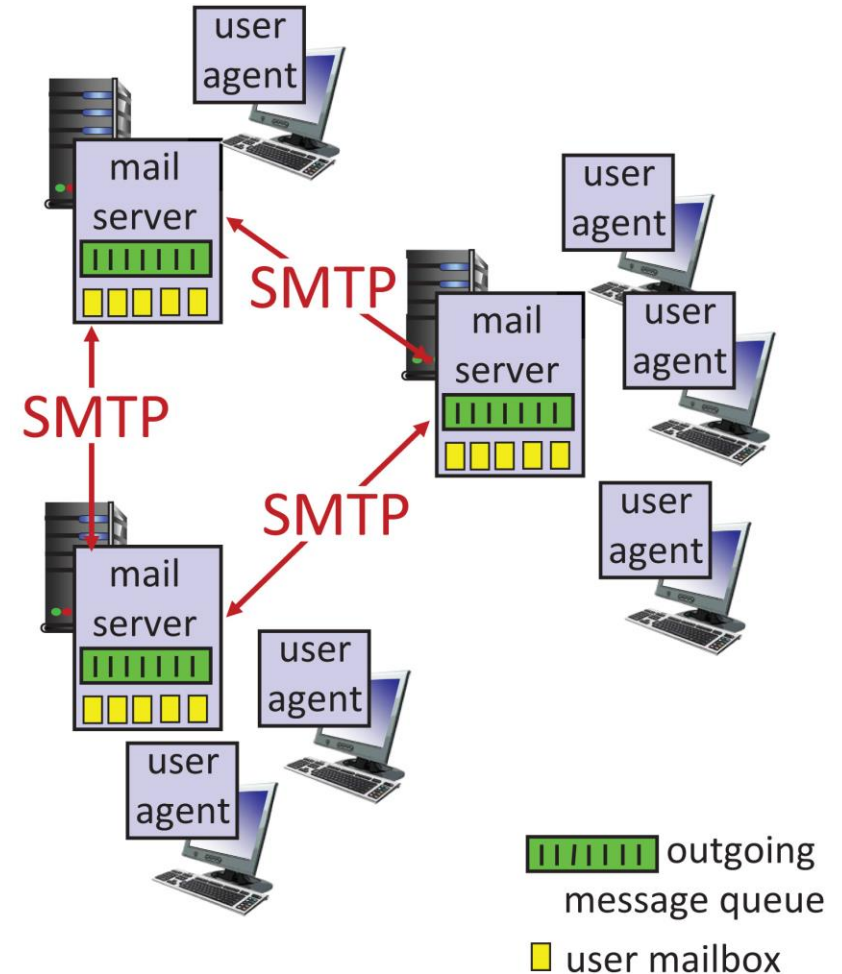
E-mail

Three major components:

- user agents
- mail servers
- simple mail transfer protocol: SMTP

User Agent

- a.k.a. “mail reader”
- composing, editing, reading mail messages
- e.g., Outlook, iPhone mail client
- outgoing, incoming messages stored on server



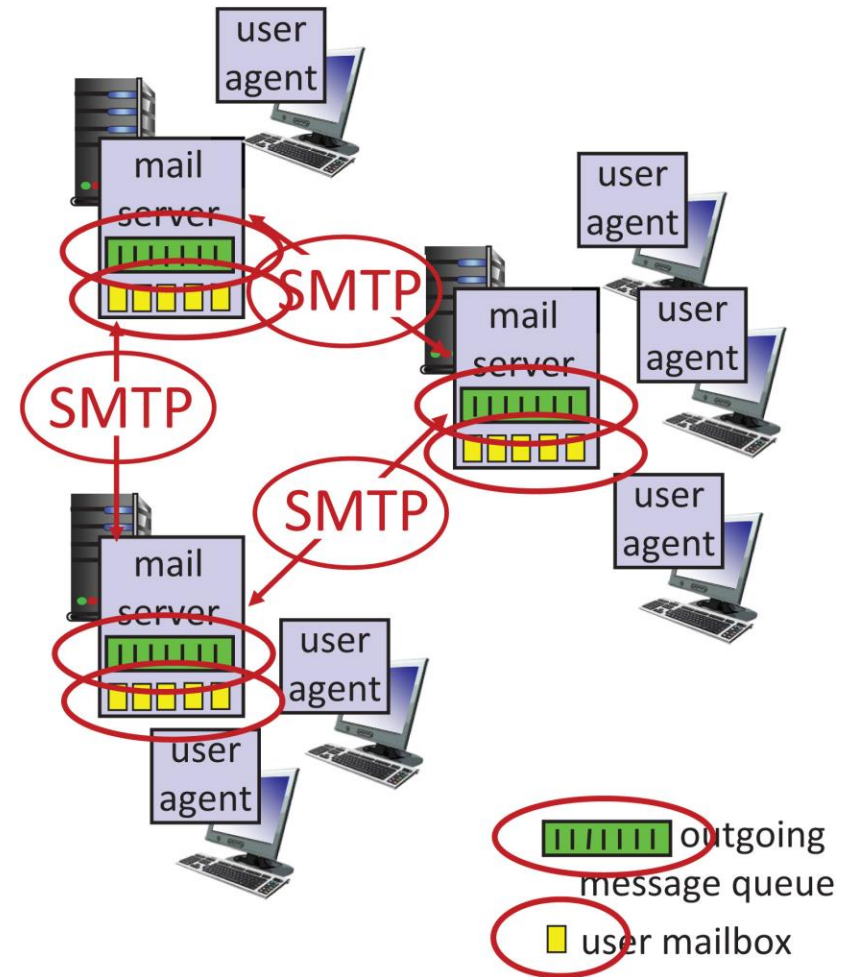
E-mail: Mail Servers

mail servers:

- **mailbox** contains incoming messages for user
- **message queue** of outgoing (to be sent) mail messages

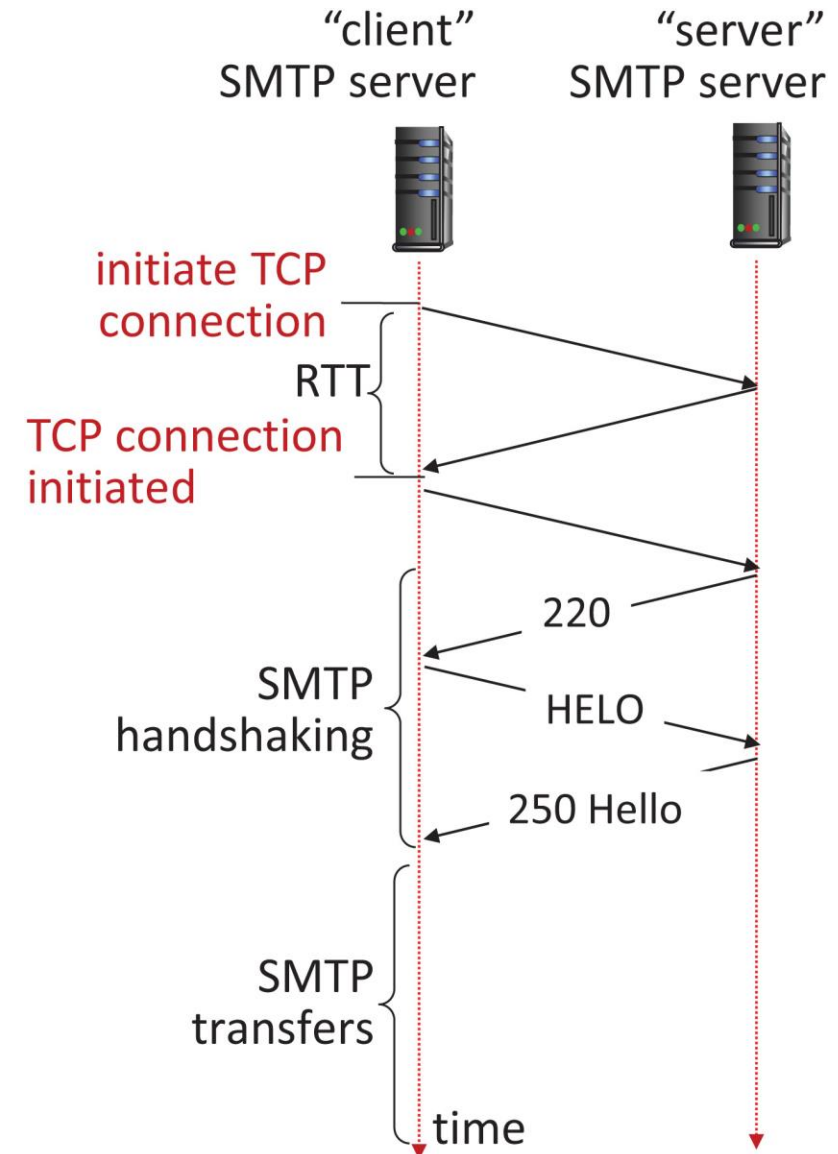
SMTP protocol between mail servers to send email messages

- **client**: sending mail server
- **“server”**: receiving mail server



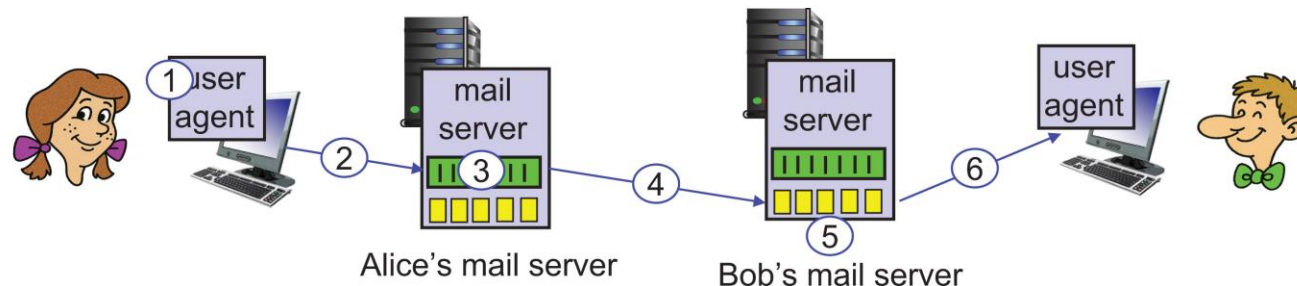
SMTP RFC (5321)

- uses TCP to reliably transfer email message from client (mail server initiating connection) to server, port 25
 - direct transfer: sending server (acting like client) to receiving server
- three phases of transfer
 - SMTP handshaking (greeting)
 - SMTP transfer of messages
 - SMTP closure
- command/response interaction (like HTTP)
 - **commands**: ASCII text
 - **response**: status code and phrase



Scenario: Alice Sends E-mail to Bob

1. Alice uses UA to compose e-mail message “to” bob@someschool.edu
2. Alice’s UA sends message to her mail server using SMTP; message placed in message queue
3. client side of SMTP at mail server opens TCP connection with Bob’s mail server
4. SMTP client sends Alice’s message over the TCP connection
5. Bob’s mail server places the message in Bob’s mailbox
6. Bob invokes his user agent to read message



Sample SMTP Interaction (1 of 2)

```
S: 220 hamburger.edu
C: HELO crepes.fr
S: 250 Hello crepes.fr, pleased to meet you
C: MAIL FROM: <alice@crepes.fr>
S: 250 alice@crepes.fr... Sender ok
C: RCPT TO: <bob@hamburger.edu>
S: 250 bob@hamburger.edu ... Recipient ok
C: DATA
S: 354 Enter mail, end with "." on a line by itself
C: Do you like ketchup?
C: How about pickles?
C: .
S: 250 Message accepted for delivery
C: QUIT
S: 221 hamburger.edu closing connection
```


SMTP: Observations

comparison with HTTP:

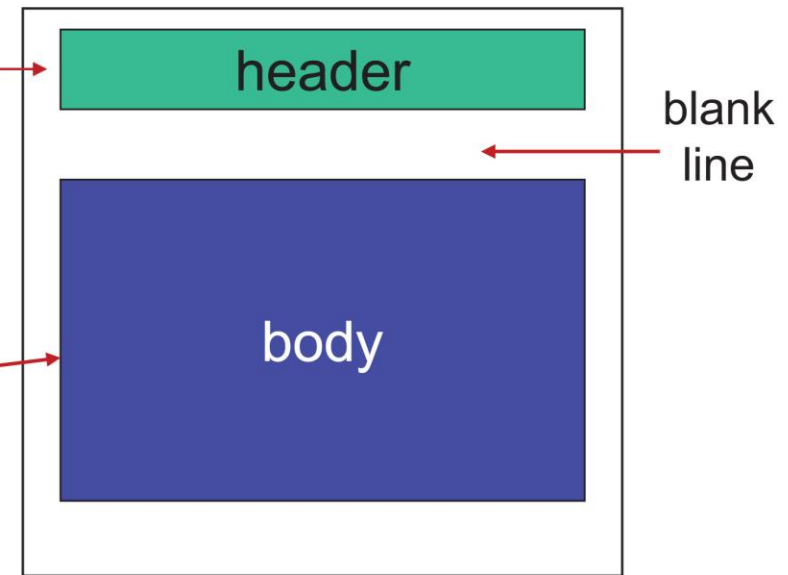
- HTTP: client pull
- SMTP: client push
- both have ASCII command/response interaction, status codes
- HTTP: each object encapsulated in its own response message
- SMTP: multiple objects sent in multipart message
- SMTP uses persistent connections
- SMTP requires message (header & body) to be in 7-bit ASCII
- SMTP server uses CRLF.CRLF to determine end of message

Mail Message Format

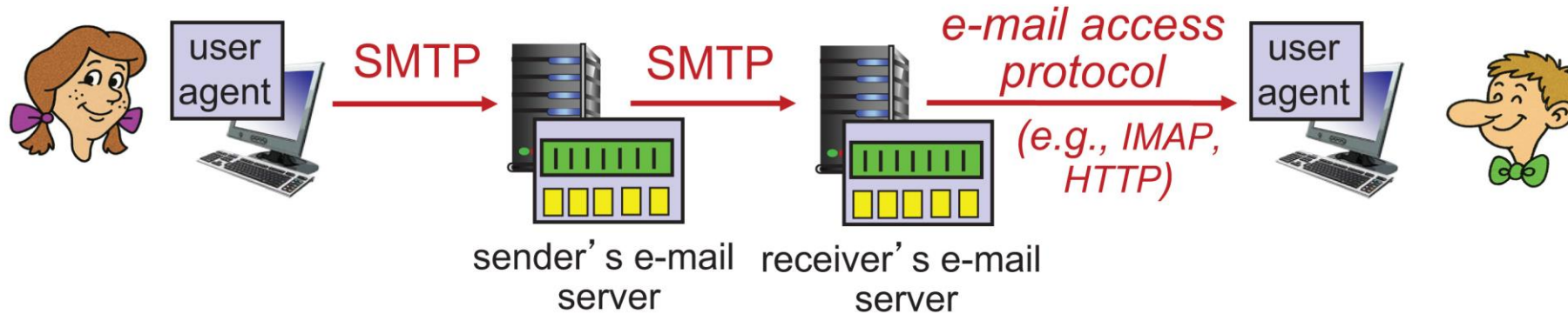
SMTP: protocol for exchanging e-mail messages, defined in RFC 5321 (like RFC 7231 defines HTTP)

RFC 2822 defines **syntax** for e-mail message itself (like HTML defines syntax for web documents)

- header lines, e.g.,
 - To:
 - From:
 - Subject:these lines, within the body of the email message area different from SMTP MAIL FROM:, RCPT TO: commands!
- Body: the “message” , ASCII characters only



Retrieving Email: Mail Access Protocols



- **SMTP**: delivery/storage of e-mail messages to receiver's server
- mail access protocol: retrieval from server
 - **IMAP**: Internet Mail Access Protocol [RFC 3501]: messages stored on server, IMAP provides retrieval, deletion, folders of stored messages on server
- **HTTP**: gmail, Hotmail, Yahoo!Mail, etc. provides web-based interface on top of SMTP (to send), IMAP (or POP) to retrieve e-mail messages

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